



ETAPA 1 - ENTENDA O PROBLEMA E ESTABELEÇA OS REQUISITOS

REQUISITOS FUNCIONAIS

- 1 - Resultado dos jogos:** Os usuários devem ser capazes de **visualizar o resultado dos jogos em tempo real e receber atualizações** sem necessidade de atualizar a página ou aplicativo
- 2 - Estatística/Histórico:** Os usuários devem ser capazes de visualizar **dados e estatísticas completas** de jogos passados, de times, campeonatos e também de jogadores específicos.

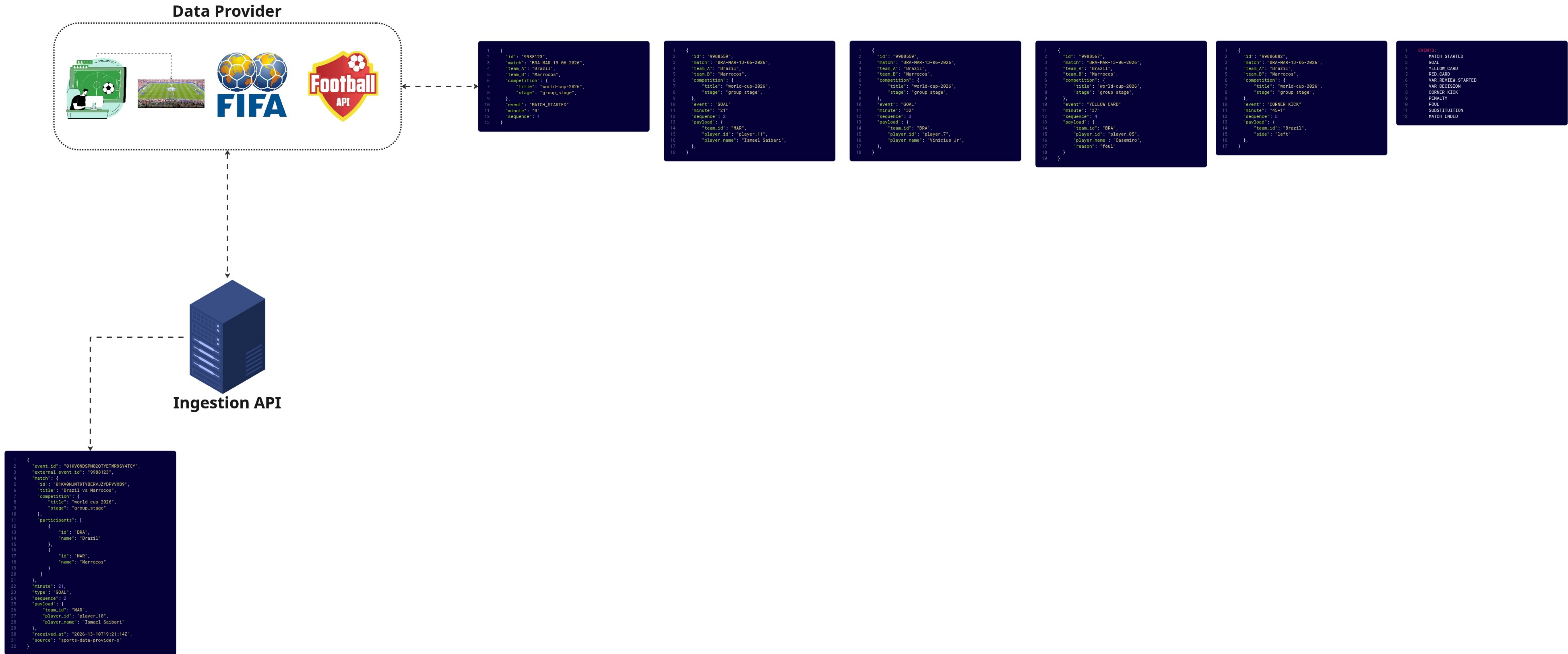
REQUISITOS NÃO FUNCIONAIS

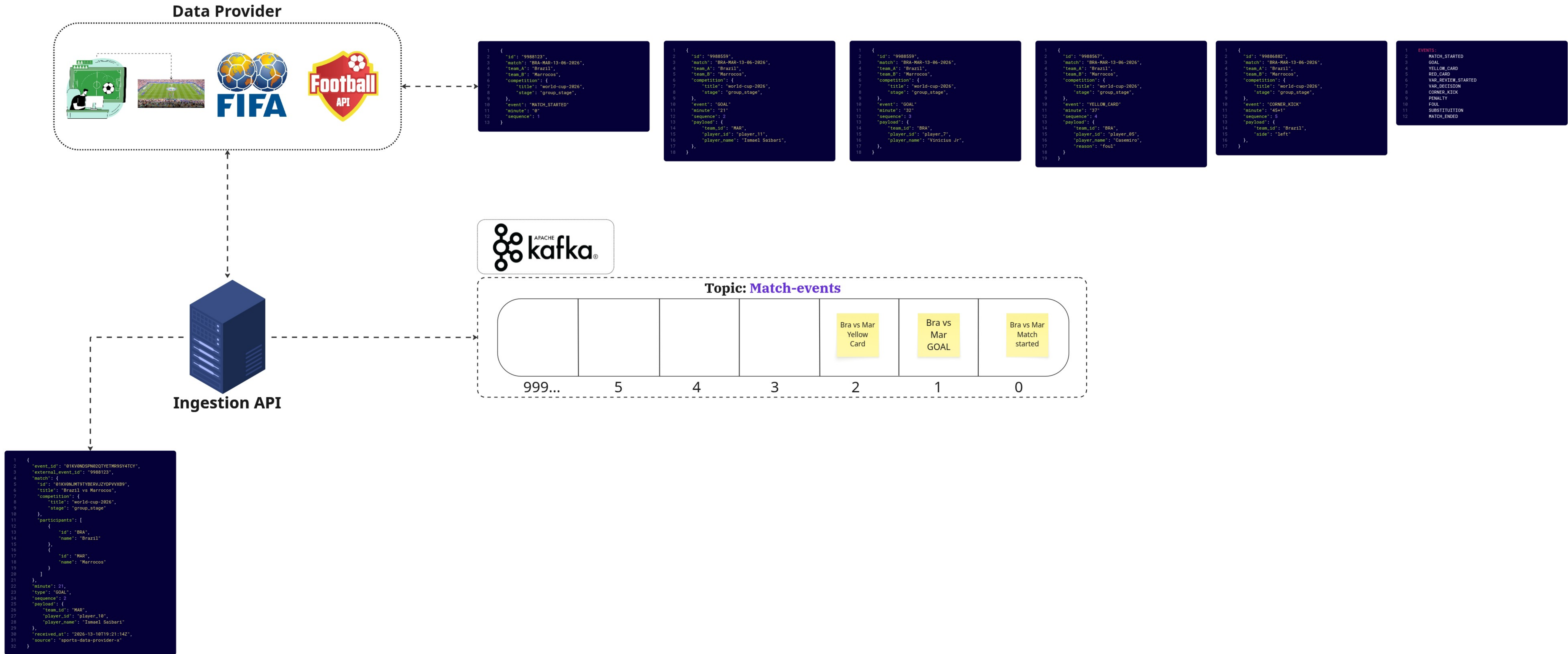
- 1 - O sistema deve suportar 10 Milhões de usuários ativos simultaneamente** nos dias de grandes jogos
- 2 - O sistema deve operar em modo de alta disponibilidade (24/7)**
- 3 - O sistema deve ter consistência forte** e não pode haver perda de dados!

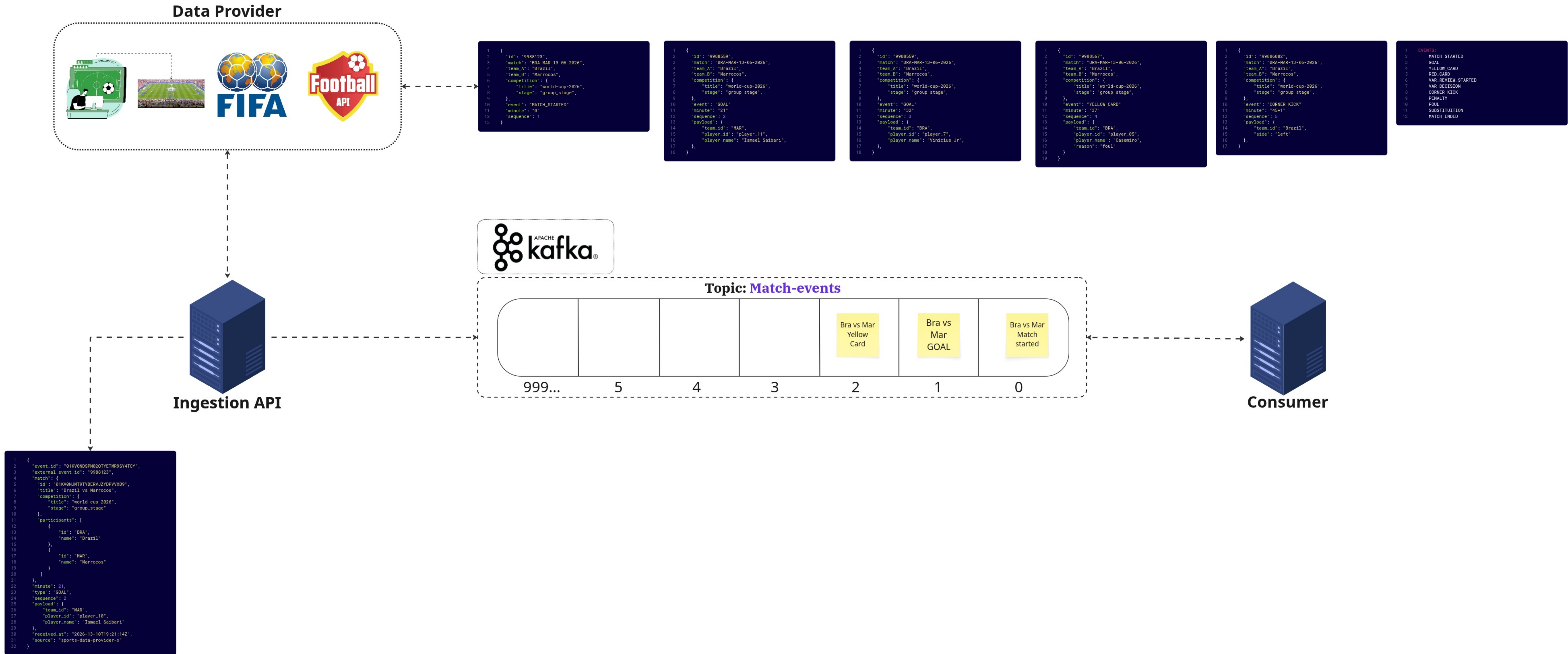


ETAPA 2 - DEFINA OS ENDPOINTS DA API

```
1 // Endpoint de acompanhamento do resultado de uma partida
2 *Request:
3 GET /matches/{match_id}
```







Data Provider



```
1 {
2   "id": "9988123",
3   "match": "BRA-MAR-13-06-2026",
4   "team_A": "Brazil",
5   "team_B": "Marrocos",
6   "competition": {
7     "title": "world-cup-2026",
8     "stage": "group_stage",
9   },
10  "event": "MATCH_STARTED",
11  "minute": "0",
12  "sequence": 1
13 }
```

```
1 {
2   "id": "9988550",
3   "match": "BRA-MAR-13-06-2026",
4   "team_A": "Brazil",
5   "team_B": "Marrocos",
6   "competition": {
7     "title": "world-cup-2026",
8     "stage": "group_stage",
9   },
10  "event": "GOAL",
11  "minute": "21",
12  "sequence": 2,
13  "payload": {
14    "team_id": "MAR",
15    "player_id": "player_10",
16    "player_name": "Ismael Saibari",
17  },
18 }
```

```
1 {
2   "id": "9988550",
3   "match": "BRA-MAR-13-06-2026",
4   "team_A": "Brazil",
5   "team_B": "Marrocos",
6   "competition": {
7     "title": "world-cup-2026",
8     "stage": "group_stage",
9   },
10  "event": "GOAL",
11  "minute": "32",
12  "sequence": 3,
13  "payload": {
14    "team_id": "BRA",
15    "player_id": "player_7",
16    "player_name": "Vinicius Jr",
17  },
18 }
```

```
1 {
2   "id": "9988567",
3   "match": "BRA-MAR-13-06-2026",
4   "team_A": "Brazil",
5   "team_B": "Marrocos",
6   "competition": {
7     "title": "world-cup-2026",
8     "stage": "group_stage",
9   },
10  "event": "YELLOW_CARD",
11  "minute": "37",
12  "sequence": 4,
13  "payload": {
14    "team_id": "BRA",
15    "player_id": "player_05",
16    "player_name": "Casemiro",
17    "reason": "foul"
18  },
19 }
```

```
1 {
2   "id": "99886882",
3   "match": "BRA-MAR-13-06-2026",
4   "team_A": "Brazil",
5   "team_B": "Marrocos",
6   "competition": {
7     "title": "world-cup-2026",
8     "stage": "group_stage",
9   },
10  "event": "CORNER_KICK",
11  "minute": "45+1",
12  "sequence": 5,
13  "payload": {
14    "team_id": "Brazil",
15    "side": "left"
16  },
17 }
```

```
1 EVENTS:
2 MATCH_STARTED
3 GOAL
4 YELLOW_CARD
5 RED_CARD
6 VAR_REVIEW_STARTED
7 VAR_DECISION
8 CORNER_KICK
9 PENALTY
10 FOUL
11 SUBSTITUTION
12 MATCH_ENDED
```



Topic: Match-events



Ingestion API

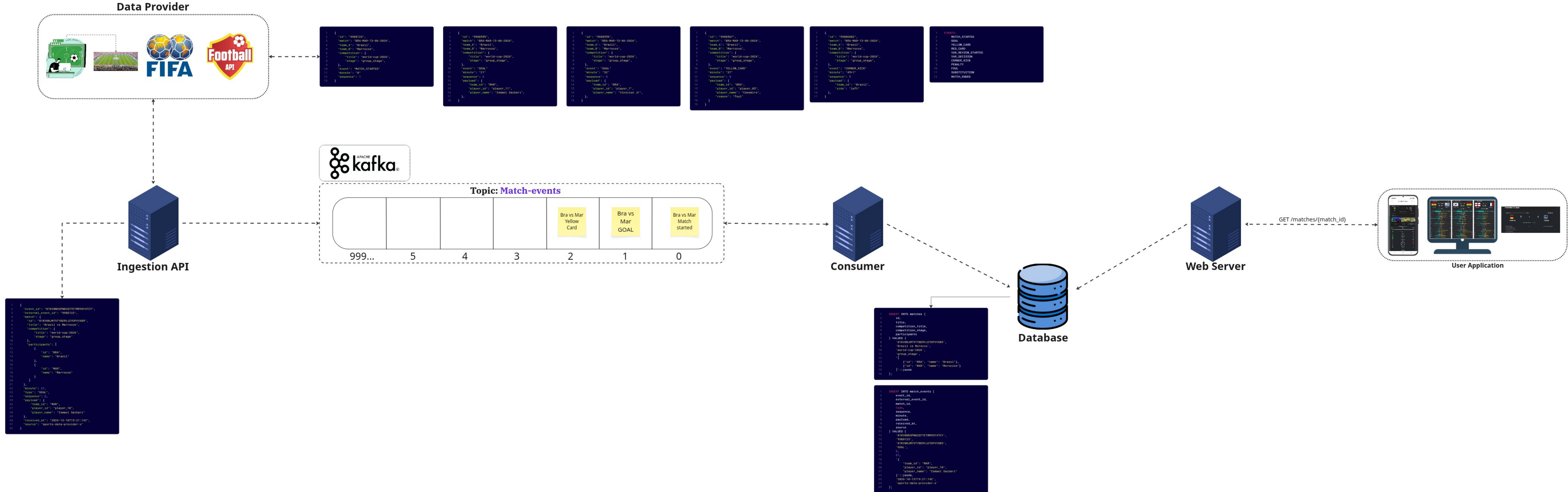
Consumer

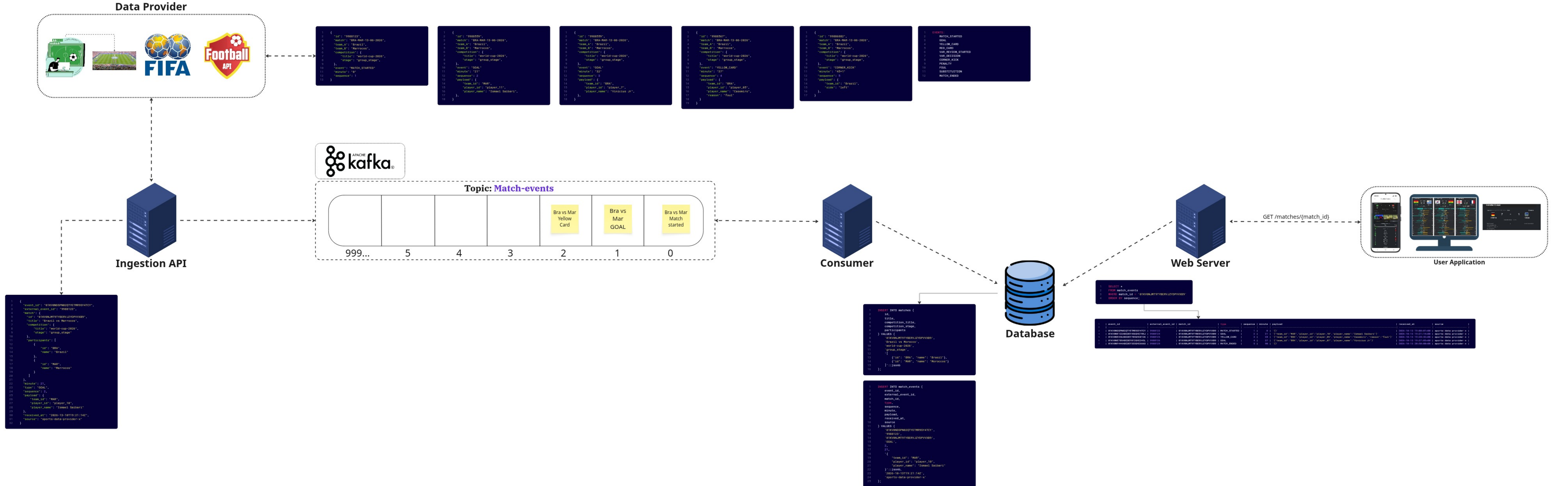
Database

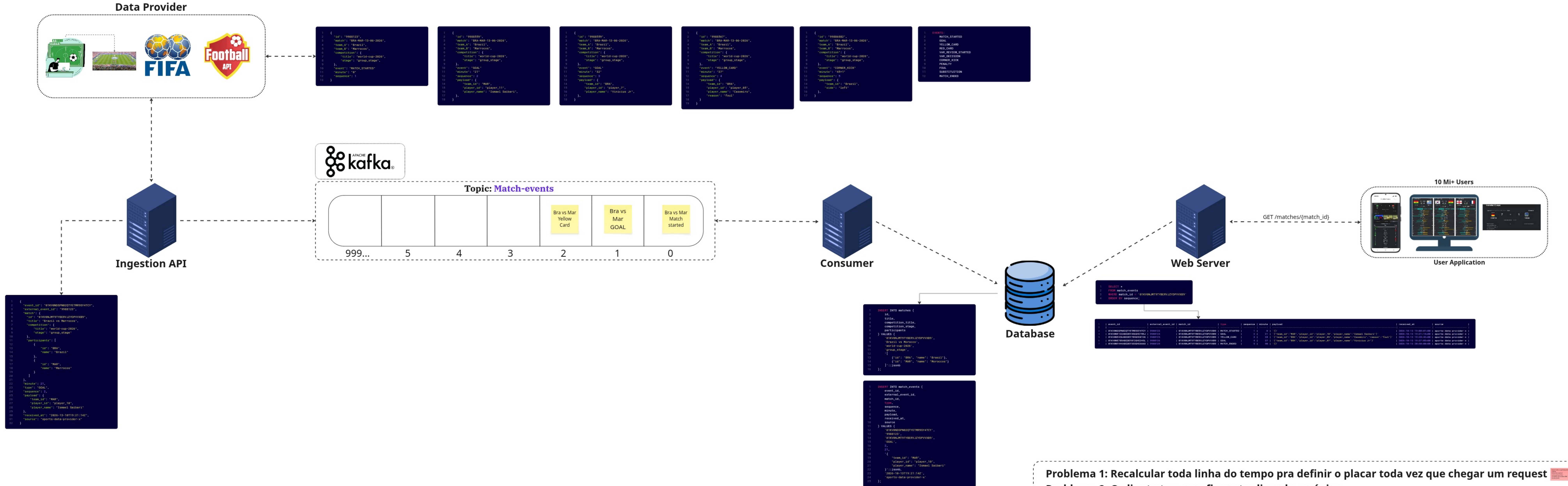
```
1 {
2   "event_id": "01KVN0SPN82QTVETMR9SY4TCY",
3   "external_event_id": "9988123",
4   "match": {
5     "id": "01KVN0JMT9TYBERVJZYDPVXXB9",
6     "title": "Brazil vs Marrocos",
7     "competition": {
8       "title": "world-cup-2026",
9       "stage": "group_stage",
10     },
11     "participants": [
12       {
13         "id": "BRA",
14         "name": "Brazil"
15       },
16       {
17         "id": "MAR",
18         "name": "Marrocos"
19       }
20     ],
21     "minute": 21,
22     "type": "GOAL",
23     "sequence": 2,
24     "payload": {
25       "team_id": "MAR",
26       "player_id": "player_10",
27       "player_name": "Ismael Saibari"
28     },
29     "received_at": "2026-10-10T19:21:14Z",
30     "source": "sports-data-provider-x"
31   }
32 }
```

```
1 INSERT INTO matches (
2   id,
3   title,
4   competition_title,
5   competition_stage,
6   participants
7 ) VALUES (
8   '01KVN0JMT9TYBERVJZYDPVXXB9',
9   'Brazil vs Morocco',
10  'world-cup-2026',
11  'group_stage',
12  [
13    {'id': 'BRA', 'name': 'Brazil'},
14    {'id': 'MAR', 'name': 'Moroccos'}
15  ]::jsonb
16 );
```

```
1 INSERT INTO match_events (
2   event_id,
3   external_event_id,
4   match_id,
5   type,
6   sequence,
7   minute,
8   payload,
9   received_at,
10  source
11 ) VALUES (
12   '01KVN0SPN82QTVETMR9SY4TCY',
13   '9988123',
14   '01KVN0JMT9TYBERVJZYDPVXXB9',
15   'GOAL',
16   2,
17   21,
18   {
19     "team_id": "MAR",
20     "player_id": "player_10",
21     "player_name": "Ismael Saibari"
22   }::jsonb,
23   '2026-10-10T19:21:14Z',
24   'sports-data-provider-x'
25 );
```

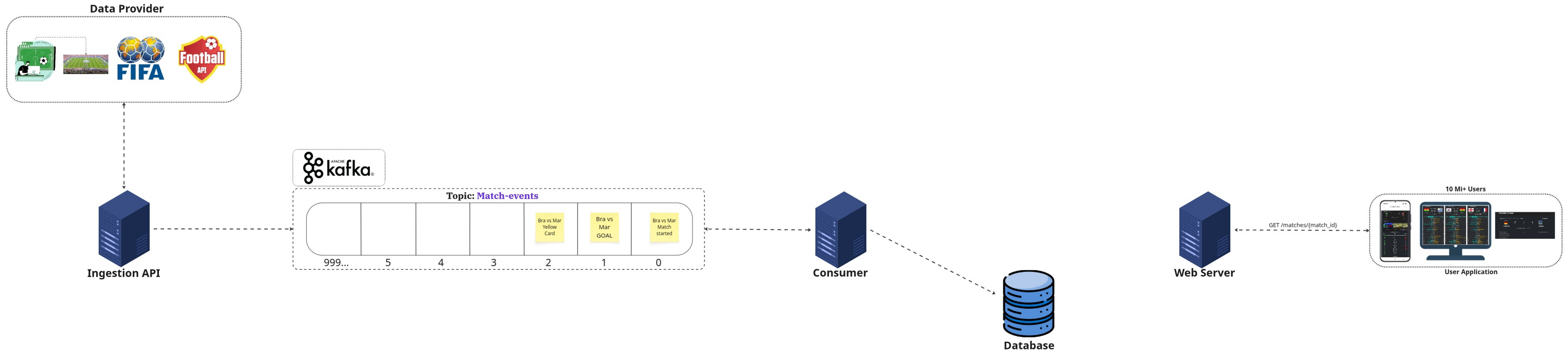


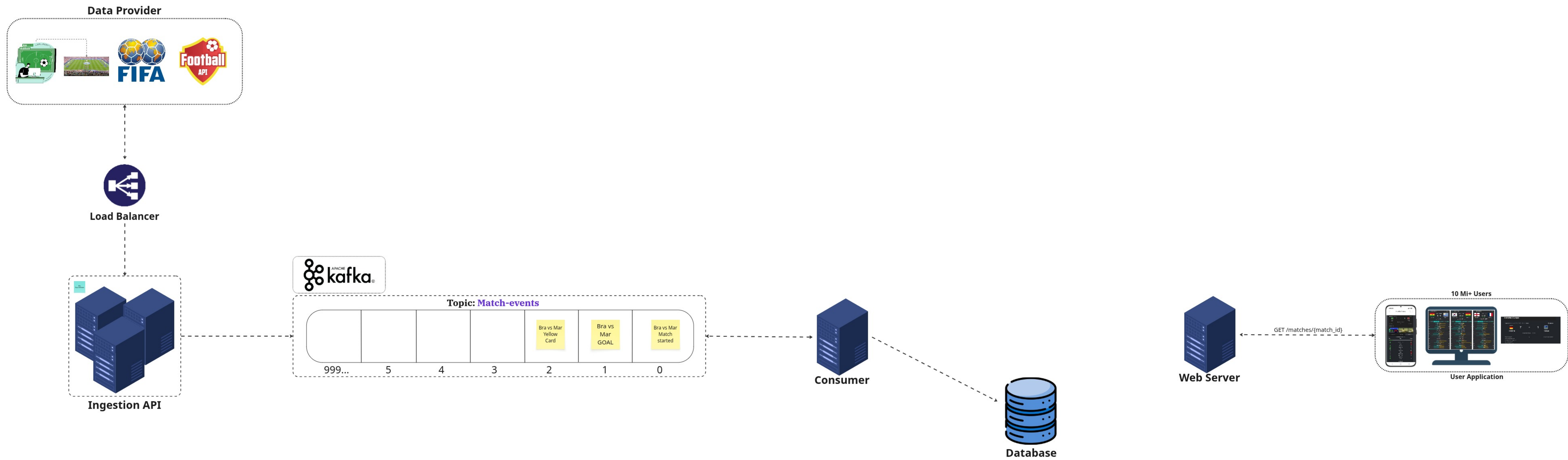


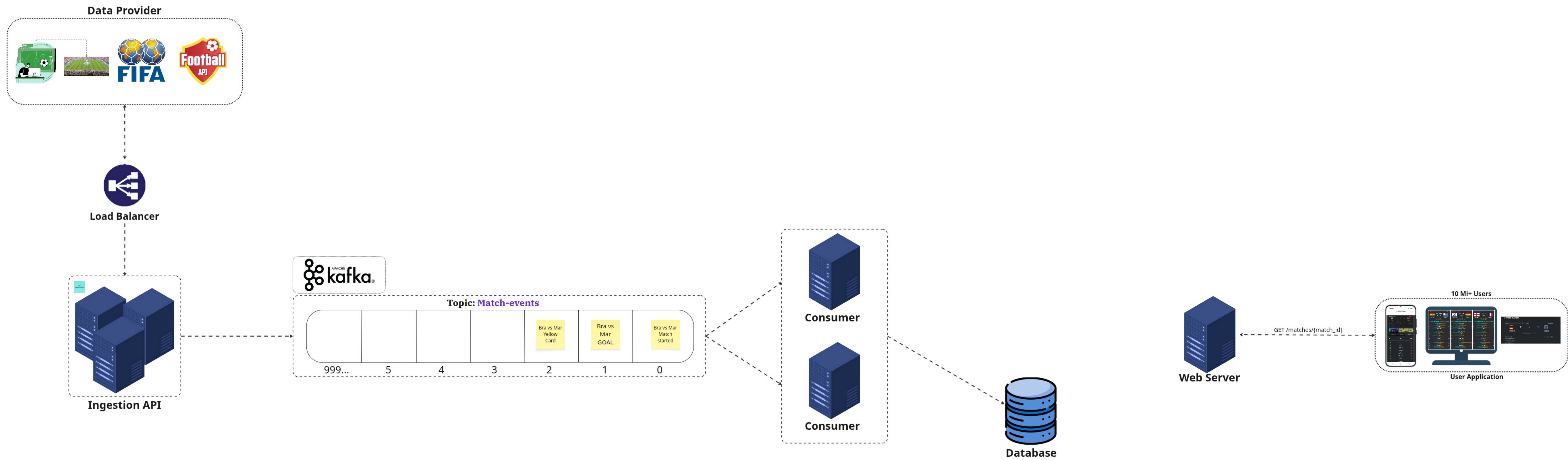


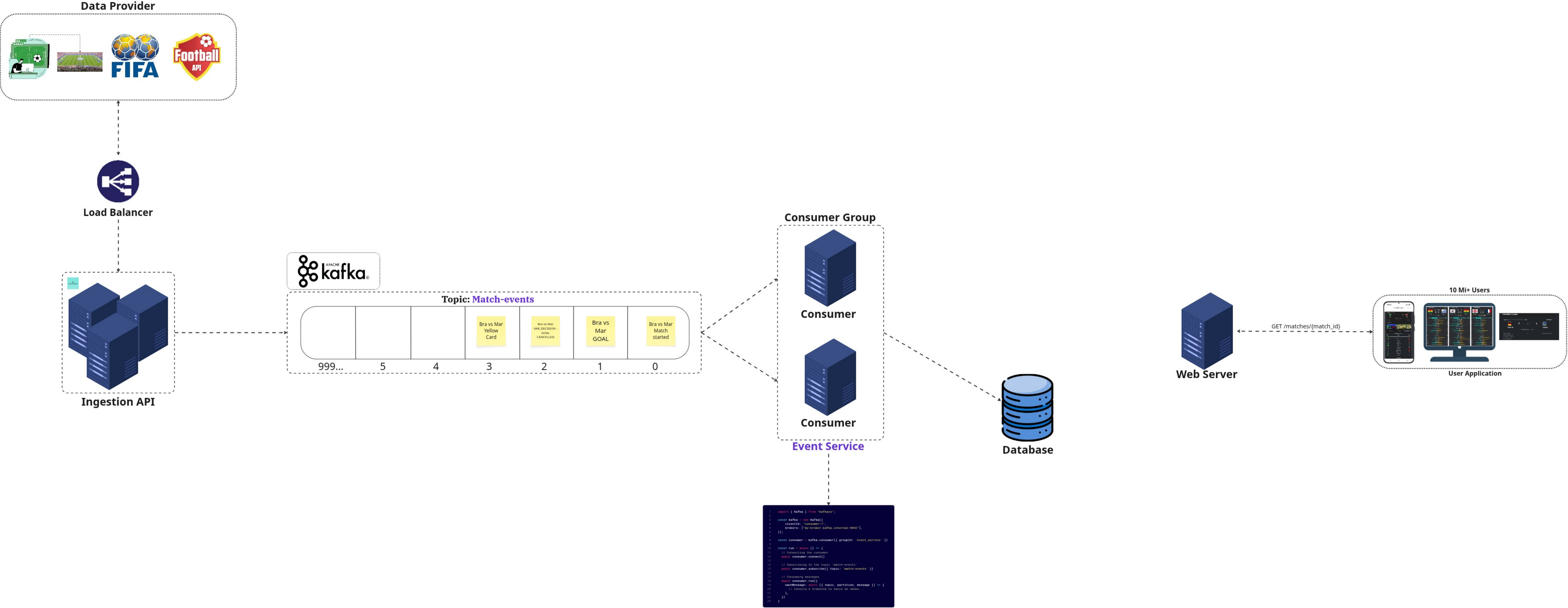
- Problema 1: Recalcular toda linha do tempo pra definir o placar toda vez que chegar um request
- Problema 2: O cliente tem que ficar atualizando a página
- Problema 3: Nossa arquitetura não tem alta disponibilidade
- Problema 4: o volume de usuários vai travar o servidor e o banco (Trends)
- Problema 5: A empresa agora decidiu atender todos os campeonatos ao redor do mundo

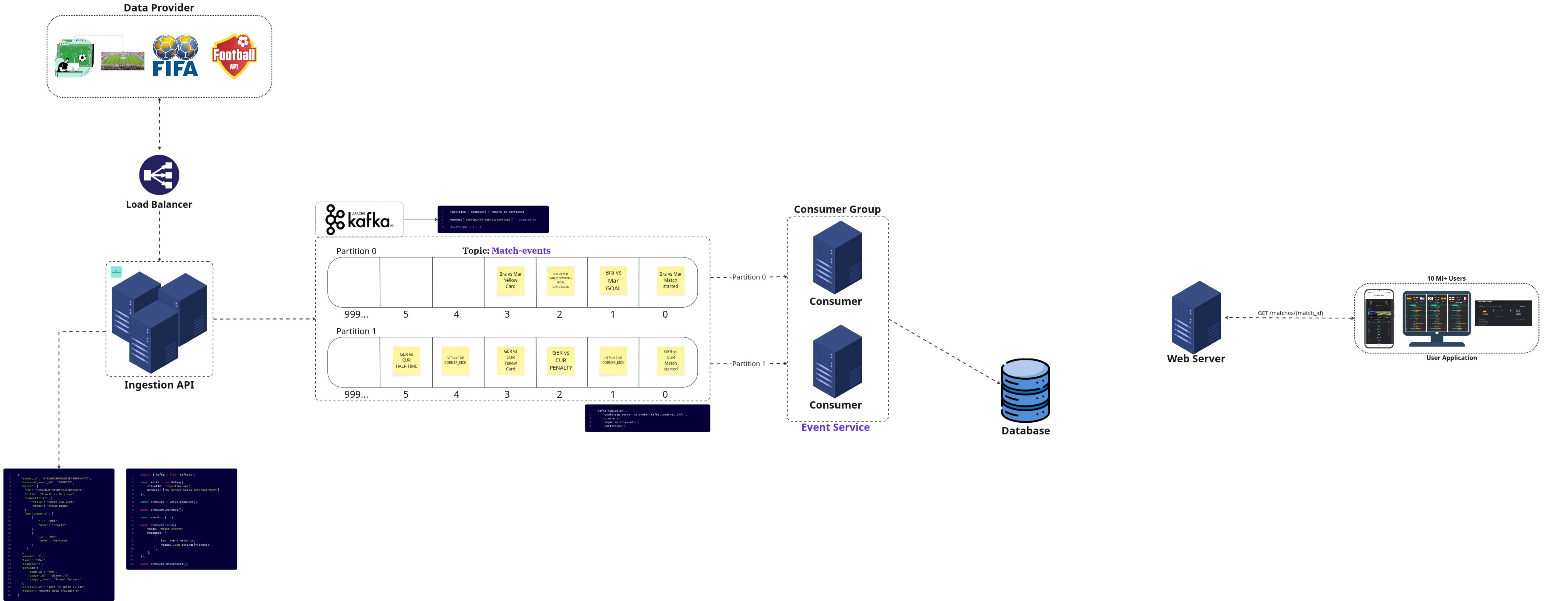


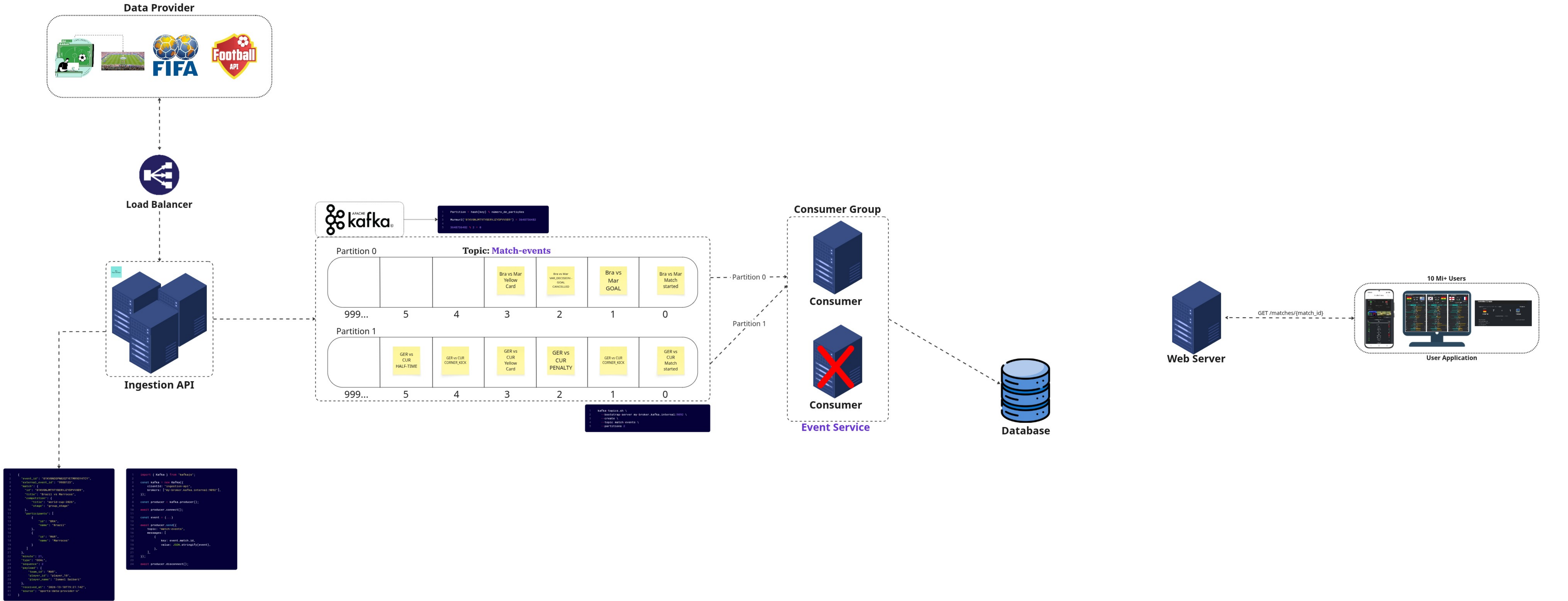


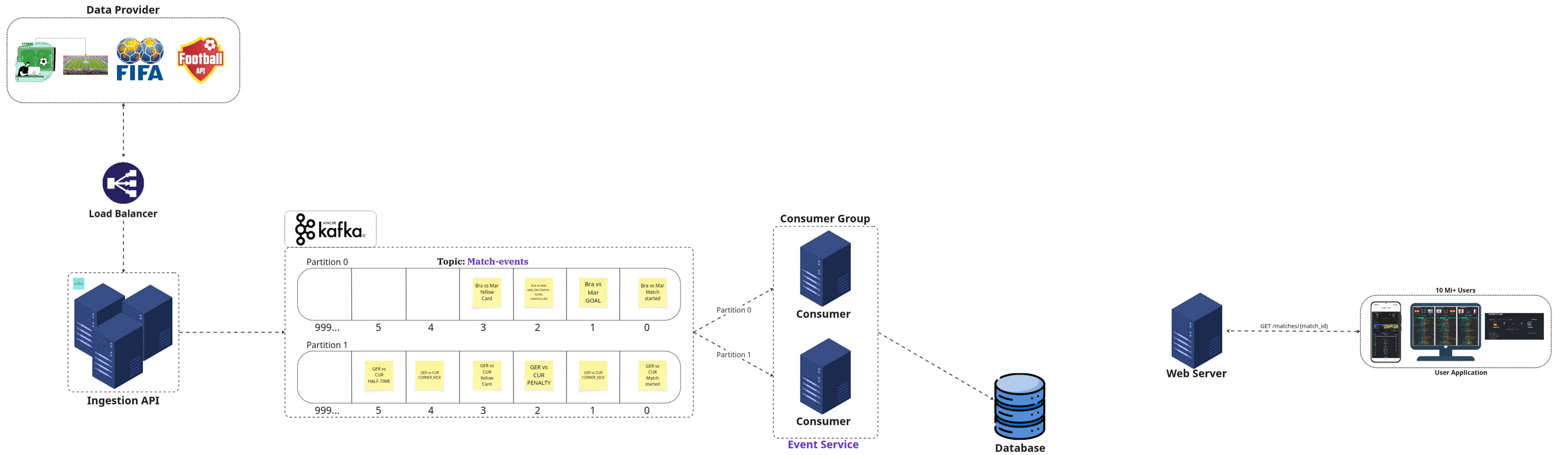


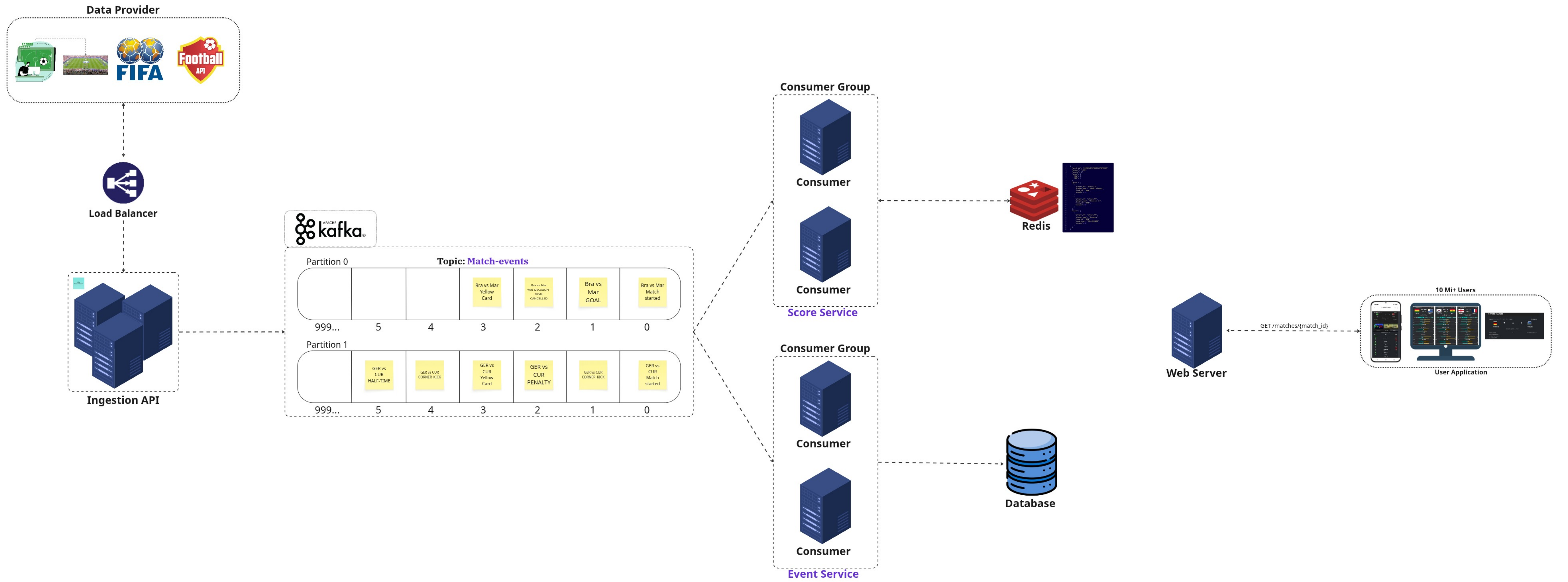


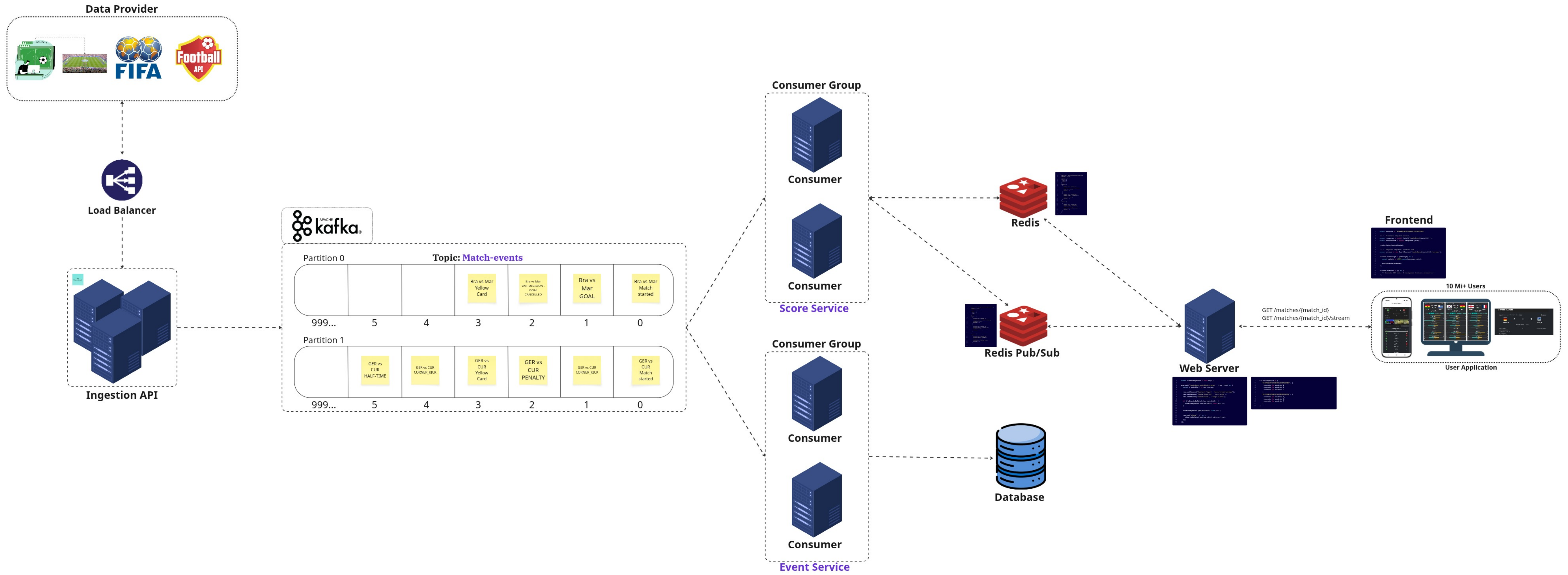


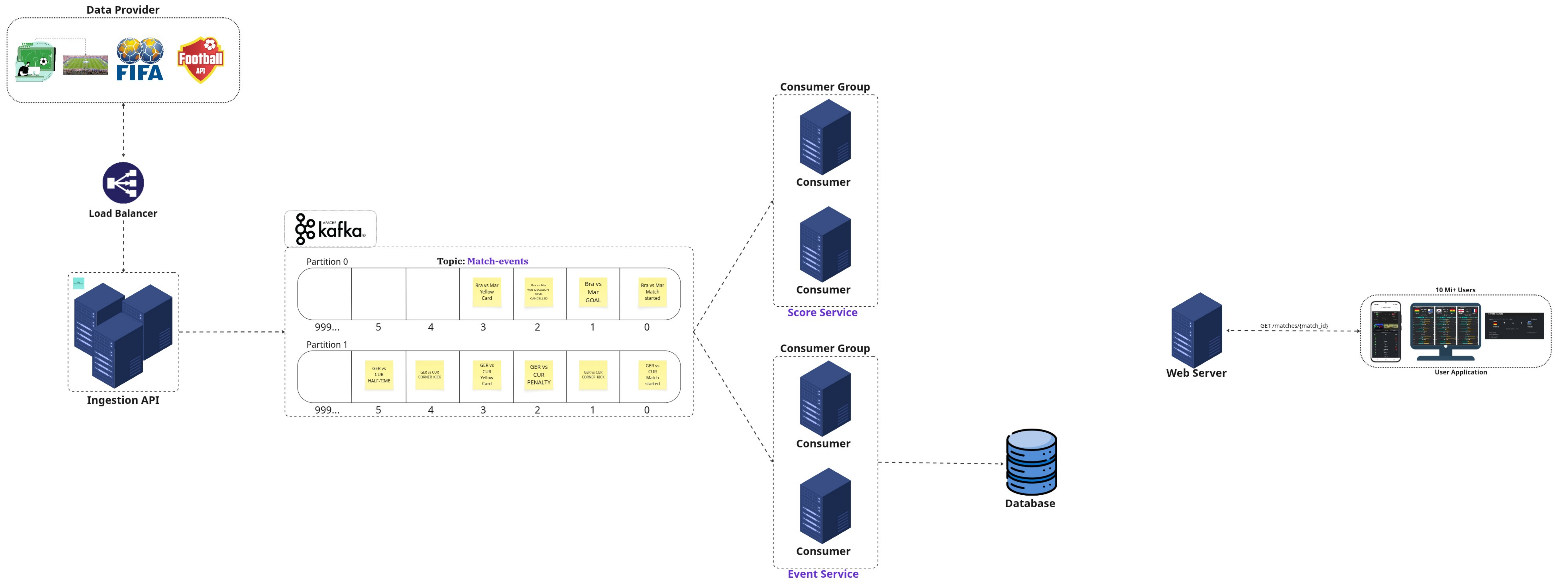


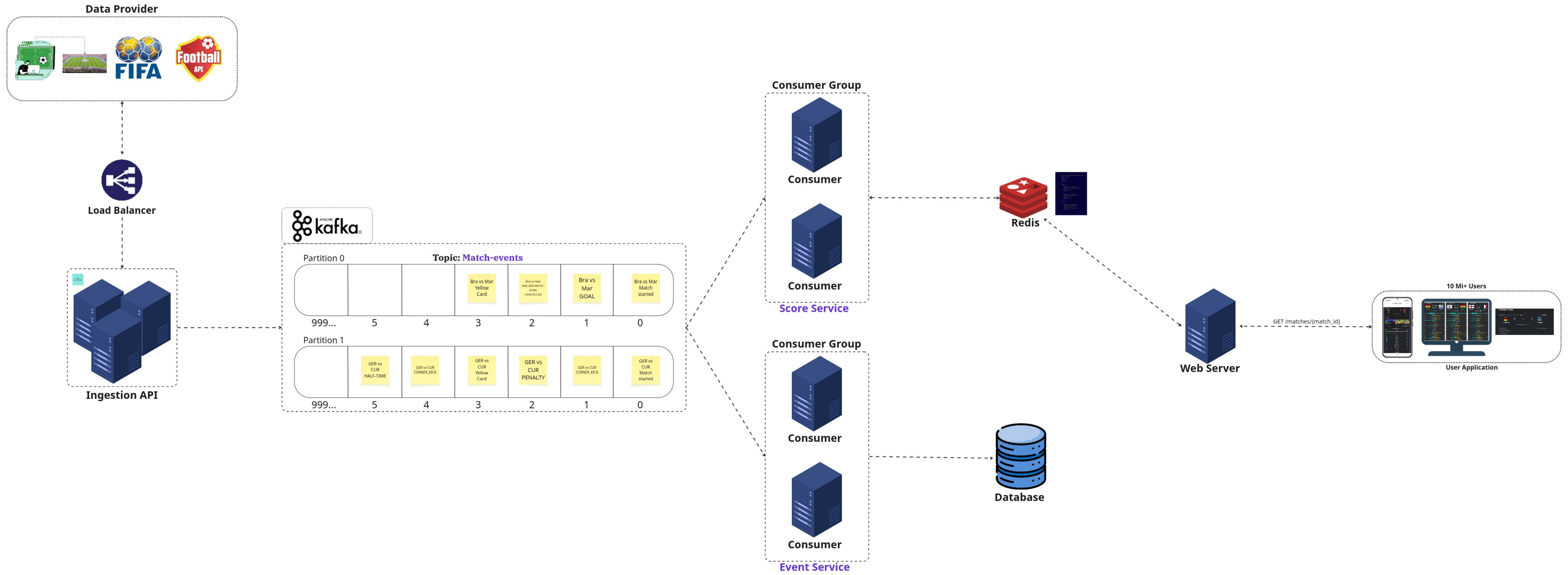


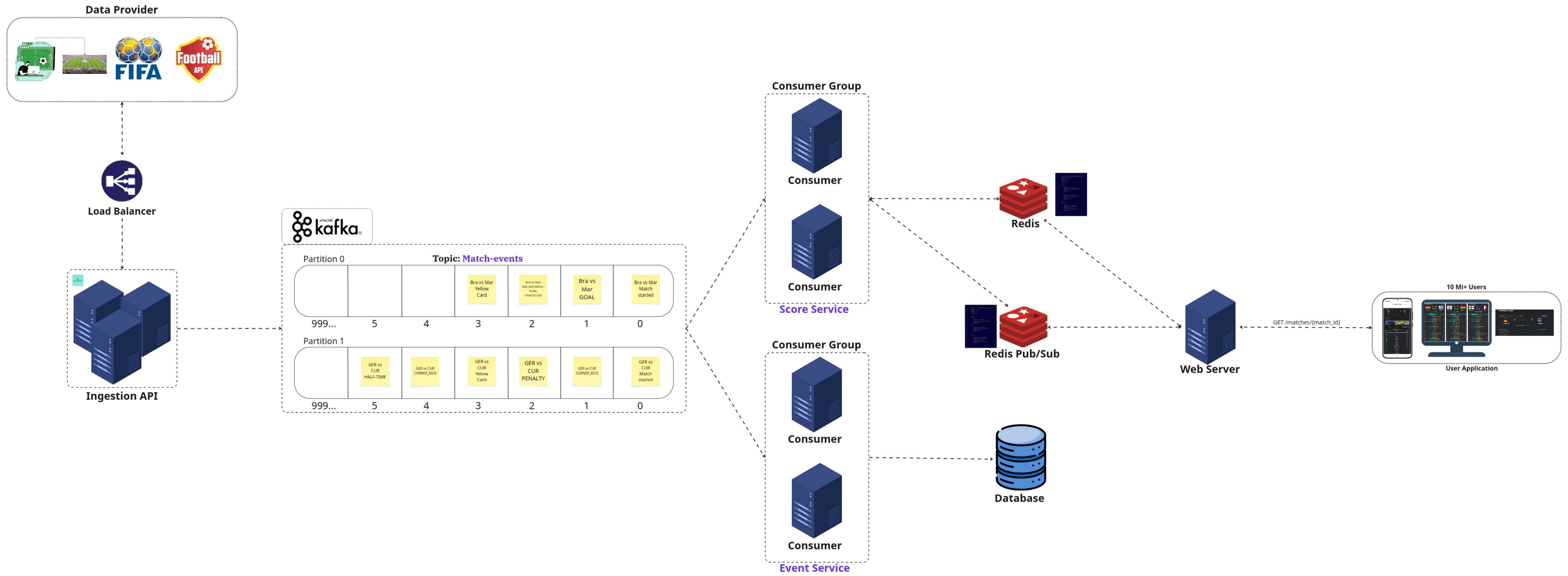


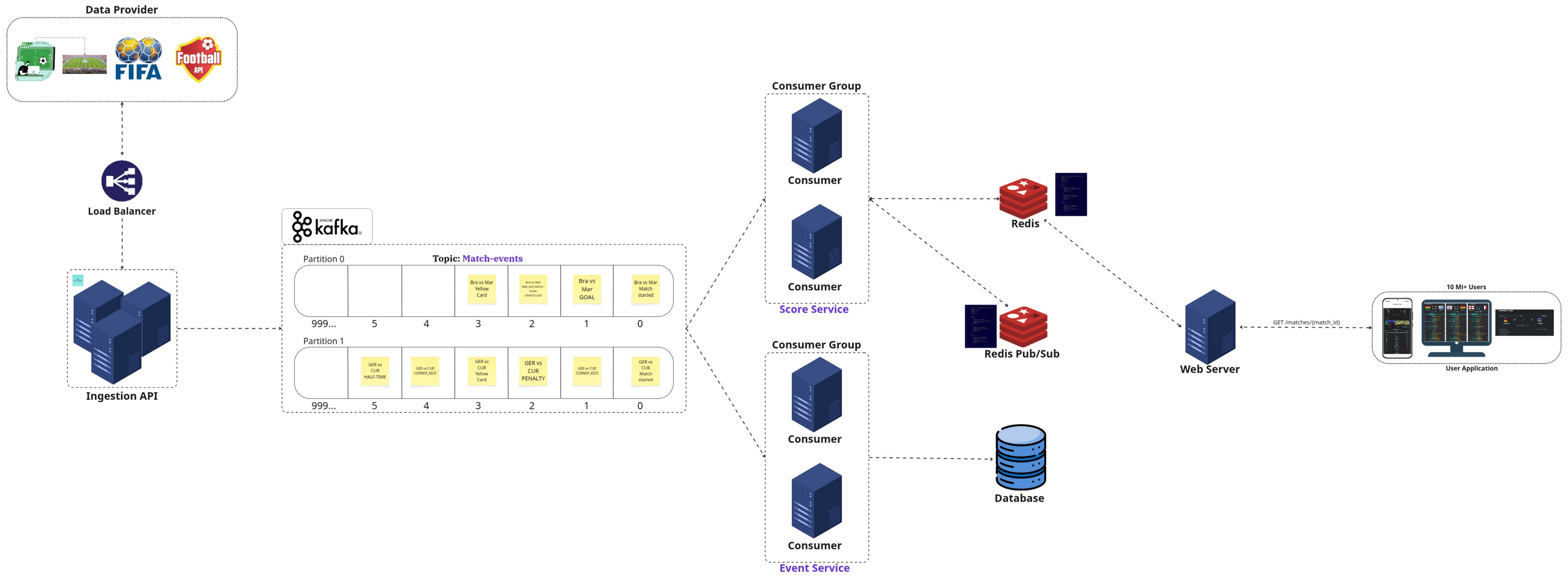


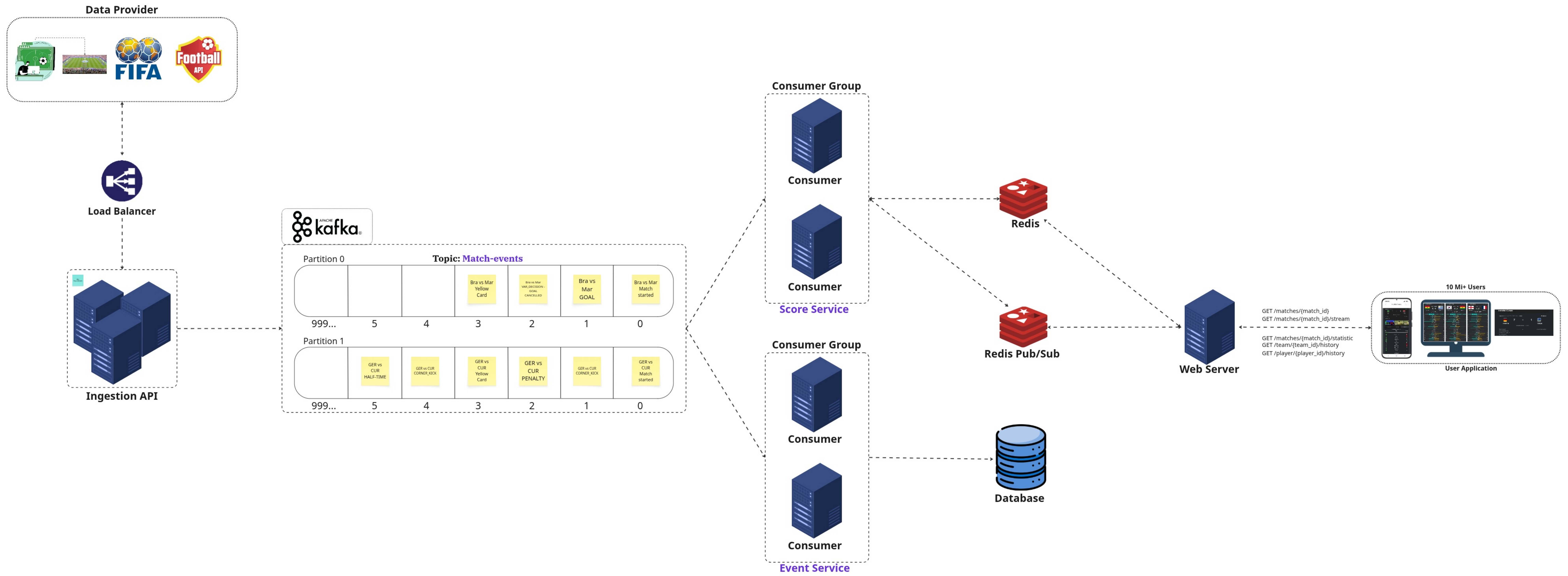


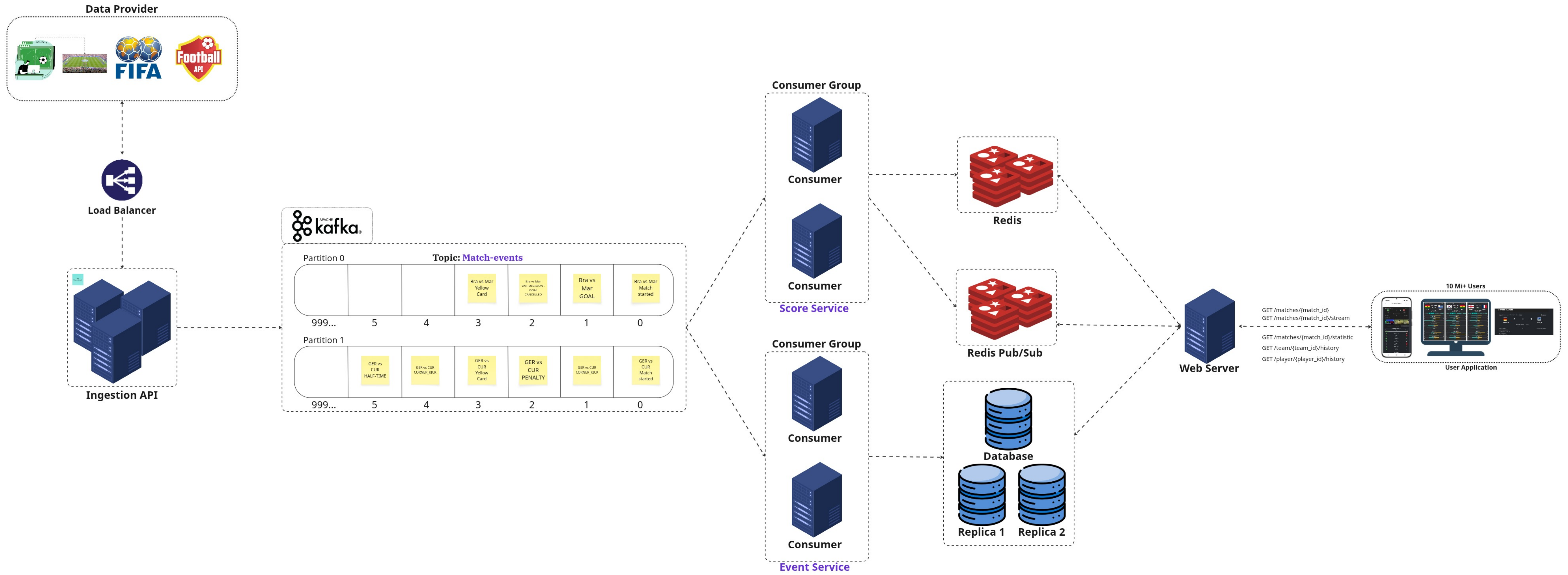


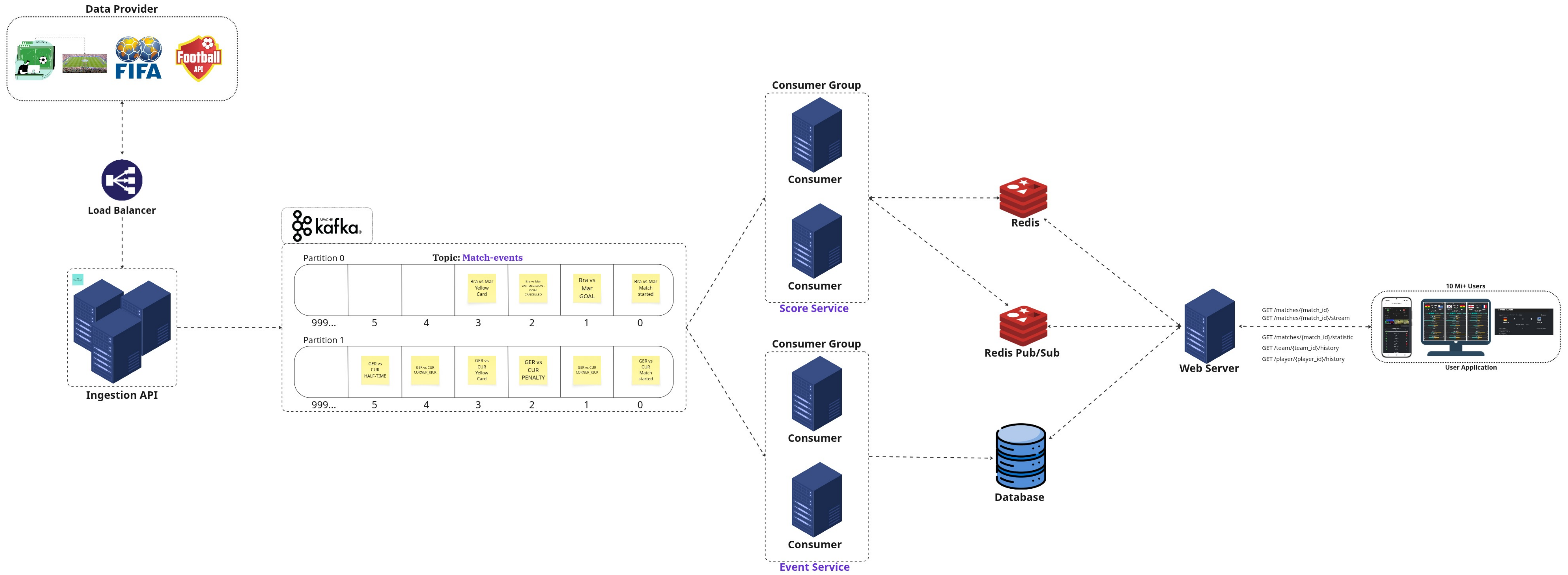












Data Provider



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5   "team_B": "Marrocos",
6   "competition": {
7     "title": "world-cup-2026",
8     "stage": "group_stage",
9   },
10  "event": "MATCH_STARTED"
11  "minute": "0"
12  "sequence": 1
13  "payload": {}
14 }
```

```
1 {
2   "id": "9988559",
3   "match": "BRA-MAR-13-06-2026",
4   "team_A": "Brazil",
5   "team_B": "Marrocos",
6   "competition": {
7     "title": "world-cup-2026",
8     "stage": "group_stage",
9   },
10  "event": "GOAL"
11  "minute": "21"
12  "sequence": 2
13  "payload": {
14    "team_id": "BRA",
15    "player_id": "player_11",
16    "player_name": "Ismael Saibari",
17  },
18 }
```

```
1 {
2   "id": "9988559",
3   "match": "BRA-MAR-13-06-2026",
4   "team_A": "Brazil",
5   "team_B": "Marrocos",
6   "competition": {
7     "title": "world-cup-2026",
8     "stage": "group_stage",
9   },
10  "event": "GOAL"
11  "minute": "32"
12  "sequence": 3
13  "payload": {
14    "team_id": "BRA",
15    "player_id": "player_7",
16    "player_name": "Vinicius Jr",
17  },
18 }
```

```
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5   "team_B": "Marrocos",
6   "competition": {
7     "title": "world-cup-2026",
8     "stage": "group_stage",
9   },
10  "event": "YELLOW_CARD"
11  "minute": "37"
12  "sequence": 4
13  "payload": {
14    "team_id": "BRA",
15    "player_id": "player_05",
16    "player_name": "Casemiro",
17    "reason": "foul"
18  },
19 }
```

```
1 {
2   "id": "99886882",
3   "match": "BRA-MAR-13-06-2026",
4   "team_A": "Brazil",
5   "team_B": "Marrocos",
6   "competition": {
7     "title": "world-cup-2026",
8     "stage": "group_stage",
9   },
10  "event": "CORNER_KICK"
11  "minute": "45+1"
12  "sequence": 5
13  "payload": {
14    "team_id": "Brazil",
15    "side": "left"
16  },
17 }
```

```
1 EVENTS:
2 MATCH_STARTED
3 GOAL
4 YELLOW_CARD
5 RED_CARD
6 VAR_REVIEW_STARTED
7 VAR_DECISION
8 CORNER_KICK
9 PENALTY
10 FOUL
11 SUBSTITUTION
12 MATCH_ENDED
```